

Solution Assignment #2 (Applied Physics)

Sol. Q#11

$$\begin{aligned} \text{(a)} \quad v_{1x} &= v_{0x} + a_{0x}(t_1 - t_0) = a_{0x} t_1 \\ &= (50 \text{ m/s}^2)(5.0 \text{ s}) = \underline{250 \text{ m/s}} \end{aligned}$$

$$\begin{aligned} \text{b)} \quad \text{Use } x_1 &= x_0 + v_{0x}(t_1 - t_0) + \frac{1}{2} a_{0x} (t_1 - t_0)^2 = \frac{1}{2} a_{0x} t_1^2 \\ &= \frac{1}{2} (50)(5) = 625 \text{ m} \end{aligned}$$

During the coasting phase, which is uniform motion with no acceleration ($a_{1x} = 0$)

$$\begin{aligned} x_2 &= x_1 + v_{1x} \Delta t \\ x_2 &= x_1 + v_{1x} (t_2 - t_1) \\ &= 625 \text{ m} + (250 \text{ m/s})(3) = 1375 \text{ m} \end{aligned}$$

$$\begin{aligned} \text{and } v_{3x}^2 &= v_{2x}^2 + 2a_{2x} \Delta x \\ v_{3x}^2 &= v_{2x}^2 + 2a_{2x} (x_3 - x_2) \end{aligned}$$

$$\text{or } \frac{v_{3x}^2 - v_{2x}^2}{2a_{2x}} = x_3 - x_2$$

$$\begin{aligned} \text{or } x_3 &= x_2 + \frac{v_{3x}^2 - v_{2x}^2}{2a_{2x}} \\ &= 1375 + \frac{0 - (250 \text{ m/s})^2}{2(-3 \text{ m/s}^2)} \end{aligned}$$

$$= 12000 \text{ m}$$

(2)

Sol. Q #2

(a)

$$V_{0y} = 14.7$$

$$V_y = 0 \quad (\text{At highest point})$$

$$\text{Use } V_y = V_{0y} - gt$$

$$\Rightarrow 0 = 14.7 - (9.8)t \Rightarrow t = \frac{14.7}{9.8}$$

$$\boxed{t = 1.50 \text{ sec}}$$

$$(b) \quad \text{Use } y - y_0 = v_{0y}t - \frac{1}{2}gt^2$$

$$y - 0 = (14.7)(1.50) - \frac{9.8}{2}(1.50)^2$$

$$\boxed{y = 11.025 \text{ m}}$$

$$c) \quad t = 2 t_{\text{high}}$$

$$\boxed{t = (2)(1.50) = 3 \text{ sec}}$$

Sol. Q #3 Displacement in 60 sec:-

$$\Delta x_1 = V_{0x}T^2/T = 480 \text{ m}$$

Displacement during the other interval of time.

$$\Delta x_2 = \int_T^\infty V_x dt = \int_T^\infty \frac{V_{0x}T^2}{t^2} dt$$

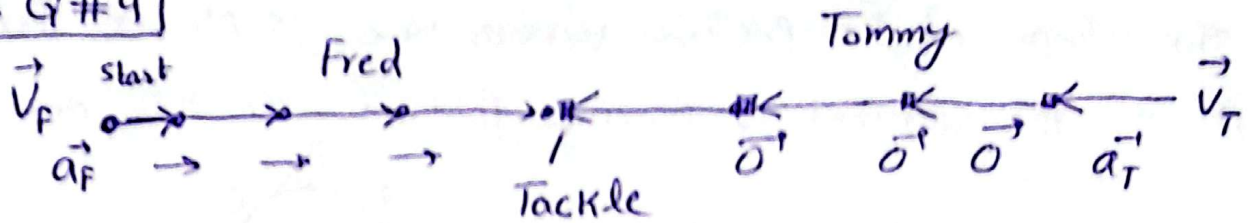
$$= V_{0x}T^2 \int_T^\infty \frac{1}{t^2} dt$$

$$= V_{0x}T^2 \left(-\frac{1}{t} \right) \Big|_T^\infty = 480 \text{ m}$$

The total displacement is

$$\Delta x = \Delta x_1 + \Delta x_2 = 960 \text{ m}$$

Sol. Q#41



$$(x_1)_F = (x_0)_F + (v_{0x})_F (t_1 - t_0) + \frac{1}{2} (a_x)_F (t_1 - t_0)^2$$

$$= \frac{1}{2} (a_x)_F t_1^2$$

$$(x_1)_T = (x_0)_T + (v_{0x})_T (t_1 - t_0) + \frac{1}{2} (a_x)_T (t_1 - t_0)^2$$

$$= (x_0)_T + (v_{0x})_T t_1$$

$$\therefore (x_1)_F = (x_1)_T$$

$$\frac{1}{2} (a_x)_F t_1^2 = (x_0)_T + (v_{0x})_T t_1$$

$$\frac{1}{2} (a_x)_F t_1^2 - (v_{0x})_T t_1 - (x_0)_T = 0$$

$$3t_1^2 + 15t_1 - 60 = 0$$

solve for t_1

$$\therefore (x_1)_F = \frac{1}{2} (a_x)_F t_1^2$$

$$= 20.6 \text{ feet}$$

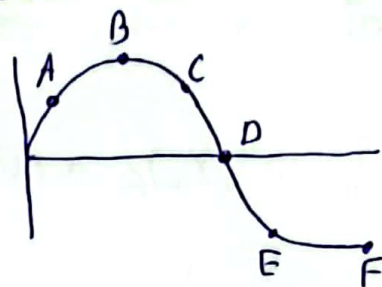
$$= 6.9 \text{ yards.}$$

Sol. Q #5

(4)

(a) The slope of the position-versus time graph is greatest at D. so the object is moving fastest at this point.

(b) The slope is negative at points C, D and E meaning the object is moving to the left at these points.



(c) At point C the slope is increasing in magnitude (getting more negative), meaning the object is speeding up to the left.

(d) At point B the object is not moving since the slope is zero. Before point B, the slope is positive, while after B it is negative, so the object is turning around at 'B'.

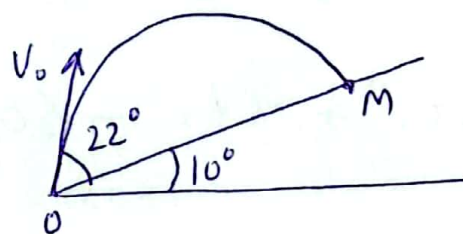
Sol. Q #6

(a) The x and y components of the displacement are given by

$$x = V_0 \cos \theta_0 t$$

$$y = V_0 \sin \theta t - \frac{1}{2} g t^2$$

$$\text{where } \theta = 22^\circ + 10^\circ = 32^\circ \quad \& \quad V_0 = 15 \text{ m/s}$$



The relationship between the coordinate x and y on the incline is given by

$$\tan(10^\circ) = \frac{y}{x}$$

Substitute x and y by their expressions above to obtain

$$\tan(10^\circ) = \frac{V_0 \sin \theta t - \frac{1}{2} g t^2}{V_0 \cos \theta t}$$

(5)

Simplify to obtain the equation in t

$$\frac{1}{2}gt + v_0 \cos \theta \tan(10^\circ) - v_0 \sin \theta = 0$$

Solve for t gives.

$$t = \frac{v_0 \sin \theta - v_0 \cos \theta \tan(10^\circ)}{0.5g}$$

$$= \frac{15 \sin(32^\circ) - 15 \cos(32^\circ) \tan 10^\circ}{0.5(9.8)}$$

$$t = 1.16 \text{ s}$$

$$b) \text{ Q. OM} = \sqrt{(v_0 \cos \theta t)^2 + (v_0 \sin \theta t - \frac{1}{2}gt^2)^2}$$

$$\text{OM} = \sqrt{(15 \cos(32^\circ)(1.16))^2 + (15 \sin(32^\circ)(1.16) - \frac{1}{2}(9.8)(1.16)^2)^2}$$

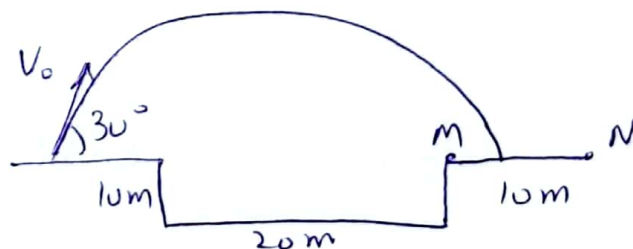
at $t = 1.16 \text{ s}$

$$\boxed{\text{OM} = 15 \text{ m}}$$

Sol. Q#7

The range is given by

$$X = \frac{v_0^2 \sin(2\theta)}{g}$$



We want to have the range greater than OM and smaller than ON, with $OM = 10 + 20 = 30 \text{ m}$

$$\text{and } ON = 10 + 20 + 10 = 40$$

$$30 < \frac{v_0^2 \sin^2(2\theta)}{g} < 40$$

$$30g / \sin(2\theta) < v_0^2 < 40g / \sin(2\theta)$$

$$\sqrt{30g / \sin(2\theta)} < v < \sqrt{40g / \sin(2\theta)}$$

$$18.4 \text{ m/s} < V_0 < 21.2 \text{ m/s}$$

Sol. Q#81 use $Y = (\tan \theta)x - \frac{1}{2} \frac{g}{(V_0 \cos \theta)^2} x^2$

hence the trajectory of a Projectile launched from ground is given by the equation

$Y = -0.025x^2 + 0.5x$, where x and Y are the coordinates of the Projectile on a rectangular sys of axes. Find the initial velocity and angle at which the Projectile is launched.

Sol use

$$Y = \tan \theta x - \frac{1}{2} \frac{g}{(V_0 \cos \theta)^2} x^2$$

hence $\tan \theta = 0.5$ which gives $\theta = \tan^{-1}(0.5)$
 $\theta = 26.5^\circ$

$$\Rightarrow -0.025 = -0.5 \left(9.8 / (V_0 \cos(26.5)) \right)^2$$

by Solving

$$V_0 = 15.6 \text{ m/s}$$

$$\theta = \tan^{-1} \left(\frac{v_{0y}}{v_{0x}} \right) = \tan^{-1} \left(\frac{9.897}{8.911} \right) = 48^\circ$$

Sol. Q #10

(a) Using $y_1 = y_0 + v_{0y}(t_1 - t_0) + \frac{1}{2} a_y(t_1 - t_0)^2$

$$h = 0 + (30 \text{ m/s}) \sin 60^\circ (4 - 0) + \frac{1}{2} (-9.8) (4 - 0)^2$$

$$h = 25.5 \text{ m}$$

b) Using $(v_y^2)_{\text{top}} = v_y^2 + 2a_y(t_{\text{top}} - y_0)$

$$0 = (v_0 \sin \theta)^2 + 2(-g) y_{\text{top}}$$

$$\Rightarrow y_{\text{top}} = \frac{(v_0 \sin \theta)^2}{2g}$$

$$y_{\text{top}} = \frac{30 \sin 60^\circ}{2(9.8)} = 34.4 \text{ m}$$

c) $v_{1y} = v_{0y} + a_y(t_1 - t_0)$

$$= v_0 \sin \theta - g t_1$$

$$= 30 \sin 60 - (9.8)(4) = -13.22 \text{ m/s}$$

$$v_{1x} = v_{0x} = v_0 \cos \theta$$

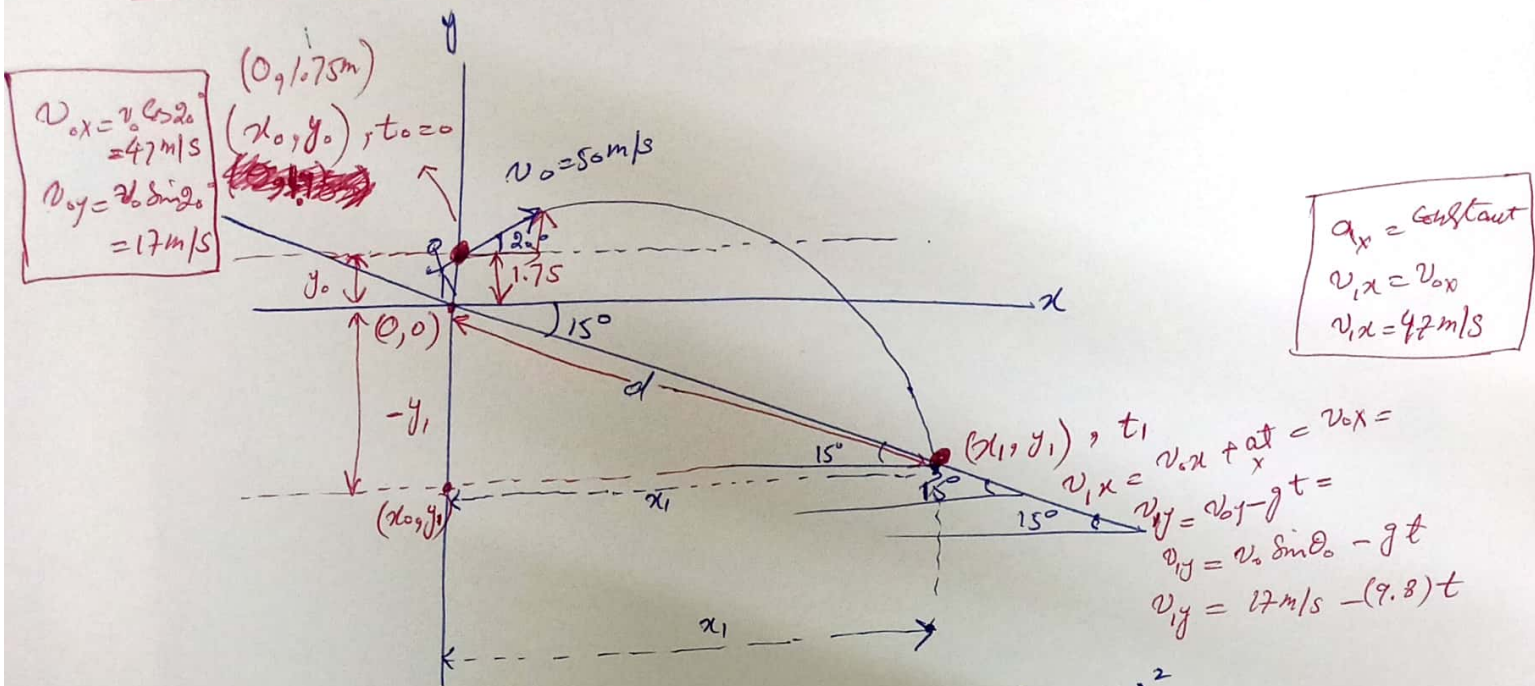
$$= 30 \cos 60^\circ = 15 \text{ m/s}$$

$$v_1 = \sqrt{v_{1x}^2 + v_{1y}^2}$$

$$\boxed{v_1 = 20 \text{ m/s}}$$

Given conditions -

The data we can extract from fig



$$\tan \theta = -y_1/x_1$$

$$y_1 = -x_1 \tan 15^\circ$$

$$y_1 = -x_1 \cdot 0.27$$

$$y_1 - y_0 = (x_1 - x_0) \tan \theta - \frac{g(x_1 - x_0)^2}{2v_0^2 \cos^2 \theta}$$

$$-0.27x_1 - 1.75 = (x_1 - 0) \tan 20^\circ - \frac{(9.8)(x_1 - 0)^2}{2(50)^2 \cos^2 20^\circ}$$

$$-0.27x_1 - 1.75 = x_1 (0.36) - \frac{(9.8)x_1^2}{2(2500)(0.94)^2}$$

$$\therefore \cos 20^\circ = 0.94$$

$$-0.27x_1 - 1.75 = 0.36x_1 - \frac{9.8x_1^2}{(5000)(0.8836)}$$

$$= 0.36x_1 - \frac{9.8x_1^2}{4400}$$

$$0.27x_1 - 1.75 = 0.36x_1 - 0.0022x_1^2$$

$$0.0022x_1^2 - 0.63x_1 - 1.75 = 0$$

$$x_1 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x_1 = \frac{+0.63 \pm \sqrt{(-0.63)^2 - 4(0.0022)(-1.75)}}{2(0.0022)}$$

$$x_1 = \frac{0.63 \pm \sqrt{0.3969 + 0.0154}}{0.0044}$$

$$x_1 = \frac{0.63 \pm \sqrt{0.4123}}{0.0044} =$$

$$x_1 = \frac{0.63 \pm 0.642}{0.0044}$$

$$x_1 = \frac{0.63 + 0.642}{0.0044}$$

$$x_1 = \frac{1.27}{0.0044}$$

$$x_1 = 288\text{ m}$$

$$x_1 = \frac{0.63 - 0.642}{0.0044}$$

$$x_1 = \frac{-0.012}{0.0044}$$

$$x_1 = -2.72\text{ m}$$

$$y = (-0.27)x_1 = (-0.27)(288)$$

$$y = -77.76\text{ m}$$

$$d = \sqrt{x^2 + y^2} = \sqrt{(288)^2 + (-77.76)^2}$$

$$d = 297\text{ m}$$